RAICES

1) Calcula:

$a)\sqrt{16}$ $b)-\sqrt{16}$ $c)\sqrt{-16}$

2) Calcula:

$a)\sqrt{\frac{4}{25}}$ $b)-\sqrt{\frac{4}{25}}$ $c)\sqrt{\frac{-4}{25}}$

3) Calcula:

$a)\sqrt[3]{\frac{8}{27}}$ $b)\sqrt[3]{\frac{-8}{27}}$ $c)-\sqrt[3]{\frac{8}{27}}$

4) Calcula:

$a)\sqrt[4]{49}$ $b)\sqrt[6]{27}$ $c)\sqrt[10]{1024}$

5) Calcula:

$a)3\sqrt{2}-5\sqrt{2}+9\sqrt{2}$ $b)(2\sqrt{2})^2$ $c)(2+\sqrt{3})^2$

6) Calcula:

$a)3\sqrt{5}-4\sqrt{5}+\sqrt{5}+2\sqrt{5}$ $b)3\sqrt{6}-2\sqrt{6}+4\sqrt{6}$

$c)\frac{2}{3}\sqrt{3}-8\sqrt{3}+\frac{4}{3}\sqrt{3}-\frac{1}{2}\sqrt{3}$ $d)6\sqrt{10}+5\sqrt{90}-3\sqrt{40}$

7) Calcula:

$a)\sqrt{3}\cdot\sqrt{2}\cdot\sqrt{5}$ $b)2\sqrt{5}\cdot 3\sqrt{2}$ $c)\sqrt{\frac{3}{4}}\cdot\sqrt{\frac{4}{3}}\cdot\sqrt{2}$ $d)\sqrt{\frac{2}{3}}\div\sqrt{\frac{2}{27}}$

8) Calcula:

$a)7\sqrt{50}-2\sqrt{32}-3\sqrt{2}-4\sqrt{18}$ $b)2\sqrt{5}-3\sqrt{45}+3\sqrt{20}$

$c)4\sqrt{18}-2\sqrt{8}-3\sqrt{32}$ $d)2\sqrt{8}-3\sqrt{28}+\sqrt{63}-2\sqrt{175}$

9) Calcula:

a) $\left(3+4\sqrt{5}\right)\cdot\left(1-2\sqrt{5}\right)$ d) $\left(4+3\sqrt{10}\right)\cdot\left(4-3\sqrt{10}\right)$

b) $\left(3+\sqrt{3}\right)\cdot\left(2-\sqrt{3}\right)$ e) $\left(2\sqrt{18}-3\right)\cdot\left(3\sqrt{18}-1\right)$

c) $\left(5+3\sqrt{6}\right)\cdot\left(7-6\sqrt{6}\right)$ f) $\left(1+\sqrt{3}\right)\cdot\left(2-\sqrt{3}\right)$

10) Racionaliza:

$a)\frac{6}{\sqrt{2}}$ $b)\frac{8}{\sqrt{5}}$ $c)\frac{6}{\sqrt{3}}$ $d)\frac{6}{\sqrt[3]{3}}$ $e)\frac{2}{\sqrt[5]{2}}$ $f)\frac{5}{\sqrt[4]{3^2}}$ $g)\frac{4}{\sqrt[7]{16}}$

11) Racionaliza:

$a)\ \frac{\sqrt{2}}{\sqrt{2}-1}$ $b)\ \frac{3\sqrt{2}}{1-\sqrt{2}}$ $c)\ \frac{3\sqrt{3}}{\sqrt{2}-\sqrt{3}}$ $d)\ \frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$ $e)\ \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

12) Opera y simplifica:

a) $\sqrt[8]{ab^3}\cdot\sqrt[6]{2a^2b^2}$

b) $\frac{\sqrt[6]{125}}{\sqrt[4]{25}}$

c) $\sqrt{x}\cdot\sqrt[3]{y}\cdot\sqrt[4]{xy}\cdot\sqrt[5]{y-x}$

d) $\frac{\sqrt[6]{(x-a)^3}}{\sqrt[4]{(x-a)^2}}$

13) Racionaliza:

a) $\frac{\sqrt{a}-\sqrt{b}}{\sqrt{ab}}$

b) $\frac{\sqrt{a-b}}{\sqrt{a+b}}$

c) $\frac{xy}{\sqrt{x}-\sqrt{y}}$

14) Opera y simplifica:

a) $\frac{\sqrt{8}-\sqrt{2}}{\sqrt{3}}=$

b) $\frac{\sqrt{27}-2\sqrt{3}}{1-\sqrt{2}}=$

c) $\frac{\sqrt[3]{3}\cdot\sqrt[3]{18}}{\sqrt{3}}=$

d) $\frac{\sqrt[3]{3}\left(\sqrt[6]{18}\right)^2}{1+\sqrt{2}}=$

e) $\frac{\sqrt[3]{16}+\sqrt[3]{54}+\sqrt[3]{2}}{\sqrt[4]{6}\cdot\sqrt[6]{\frac{36}{9}}}=$

15) Opera y simplifica:

a) $\frac{x+4+4\sqrt{x}}{2+\sqrt{x}}$

b) $\sqrt{\frac{ab^2+a^2b}{a+b}+\sqrt{\frac{a^2b-ab^2}{a-b}}}$

c) $\frac{\sqrt{3}-2}{5+2\sqrt{3}}-\frac{1}{\sqrt{3}+1}+\frac{1}{\sqrt{3}}$

d) $\sqrt{\frac{1+2\sqrt{5}}{1-2\sqrt{5}}}+\sqrt{\frac{1-2\sqrt{5}}{1+2\sqrt{5}}}$

16) Calcula: $\sqrt{\sqrt[3]{\sqrt{\sqrt[3]{3\sqrt{3}}}}}$